

PRACTICAL 7

Code & Output

```
data = c(8.35 ,9.06, 7.77, 8.49, 9.19, 10.20, 11.68, 12.86, 12.69, 12.41, 11.67, 9.44, 6.99, 6.51, 6.20,
6.96, 8.94, 10.04, 11.04, 12.17, 12.88, 14.40, 13.74, 11.77, 9.72, 6.44, 3.93, 4.47, 7.47, 9.81, 11.39,
12.19, 13.01, 12.06, 10.39, 7.17, 6.81, 7.71, 9.76, 10.82, 12.01, 12.04, 11.14, 10.85, 10.21, 8.54, 8.49,
10.13, 11.32, 12.86, 12.80, 12.87, 12.80, 12.95, 12.58, 12.83, 11.88, 11.95, 10.70, 9.43 ,7.28, 7.27,
8.38, 9.90, 10.09, 11.45, 11.45, 7.97, 7.18 ,5.79, 8.60, 9.44, 8.65, 9.56, 9.98, 10.36, 11.92, 13.49,
13.34, 12.03, 10.20, 8.30, 8.16 ,7.73, 7.77, 8.54, 9.70, 12.59, 13.89, 14.42, 11.97, 10.82, 12.05, 13.23,
12.92 ,13.56, 12.87, 10.27, 6.67,6.20)
mat = matrix(rep(0,10),nrow=2,dimnames=list(c("AR","MA"),1:5))
for(i in 1:5)
{
  ar_est=arima(data,order=c(i,0,0))
  mat[1,i]=ar_est$aic
  ma_est=arima(data,order=c(0,0,i))
  mat[2,i]=ma_est$aic
}
mat
```

	1	2	3	4	5
AR	353.9770	295.2669	295.4384	297.3666	299.0679
MA	371.6931	328.9529	311.2658	308.3583	308.6378

Best model

```
best=arima(data,order=c(2,0,0))
best
```

Call:

```
arima(x = data, order = c(2, 0, 0))
```

Coefficients:

	ar1	ar2	intercept
	1.3708	-0.6784	10.2010
s.e.	0.0725	0.0728	0.3273

sigma^2 estimated as 1.012: log likelihood = -143.63, aic = 295.27

Normality of residuals

```
res=residuals(best)
qqnorm(res,main="Q-Q plot of residuals")
qqline(res)
```

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Group : B

```
shapiro.test(res)
```

Shapiro-Wilk normality test

data: res

W = 0.98556, p-value = 0.3483

Time series plot along with fitted time series plot

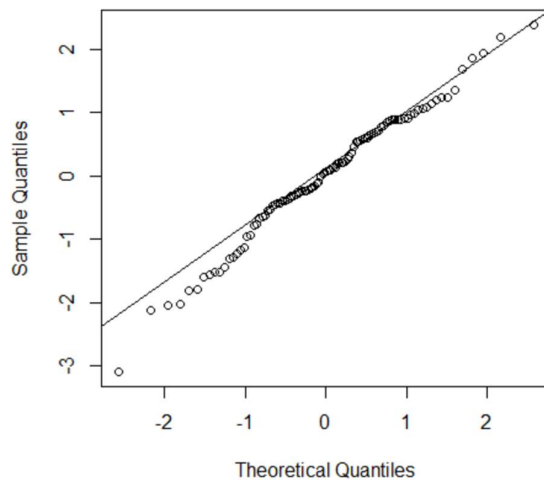
```
y_fit=data-res
```

```
ts.plot(data,main="Time series plot of yt")
```

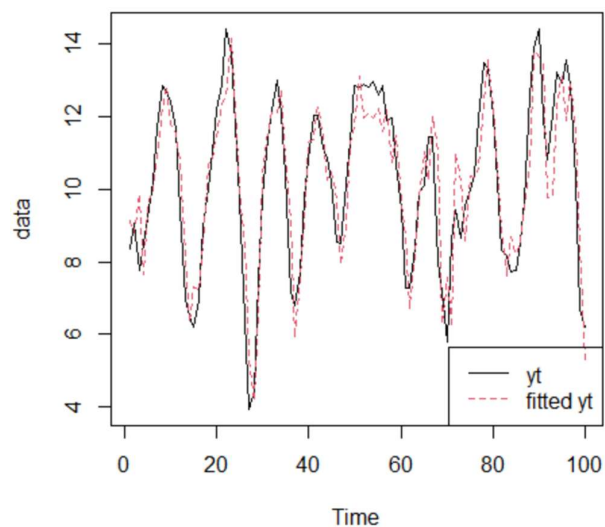
```
points(y_fit,type="l",col=2,lty=2)
```

```
legend("bottomright",lty=1:2,col=1:2,legend=c("yt","fitted yt"))
```

Q-Q plot of residuals



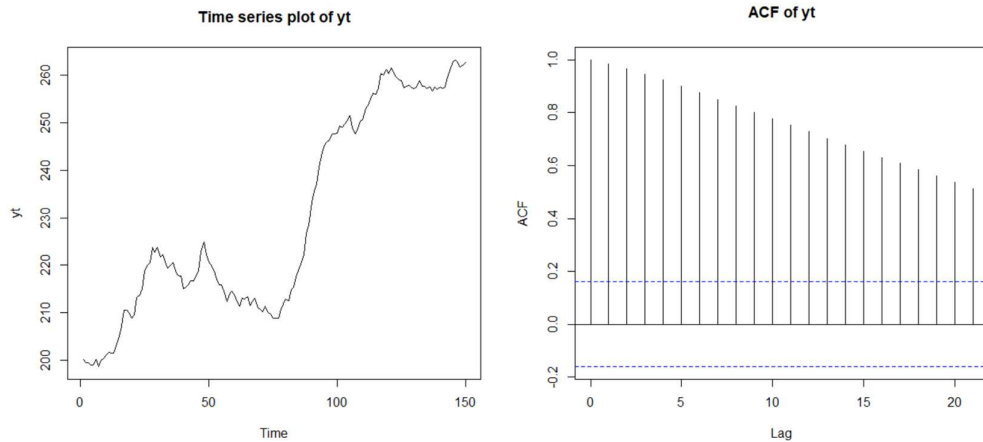
Time series plot of yt



PRACTICAL 8

Code & Output

```
library(tseries)
par(mfrow=c(1,2))
yt=BJsales
plot(yt,main="Time series plot of yt")
acf(yt,main="ACF of yt")
```



```
adf.test(yt)
```

Augmented Dickey-Fuller Test

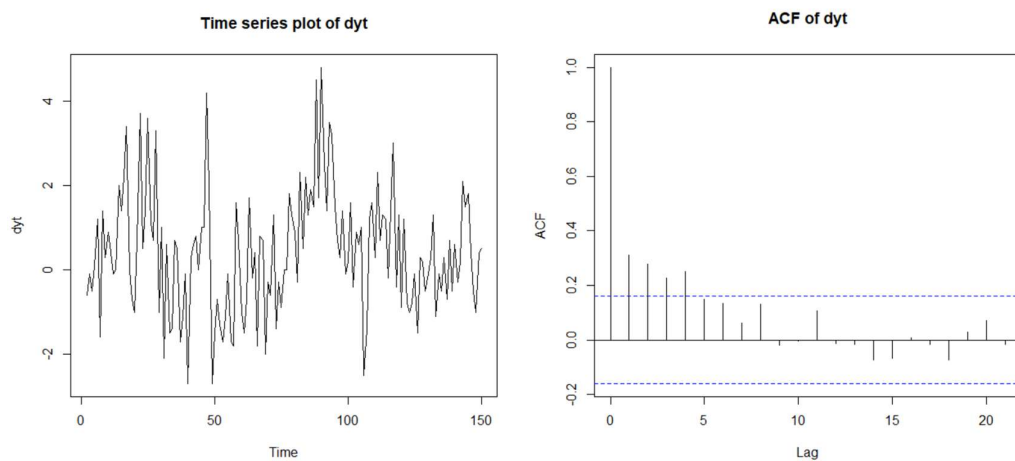
data: yt

Dickey-Fuller = -2.1109, Lag order = 5, p-value = 0.5302

alternative hypothesis: stationary

After 1 differencing

```
dyt = diff(yt)
plot(dyt,main="Time series plot of dyt")
acf(dyt,main="ACF of dyt")
```



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```
adf.test(dyt)
```

Augmented Dickey-Fuller Test

data: dyt

Dickey-Fuller = -3.3485, Lag order = 5, p-value = 0.06585

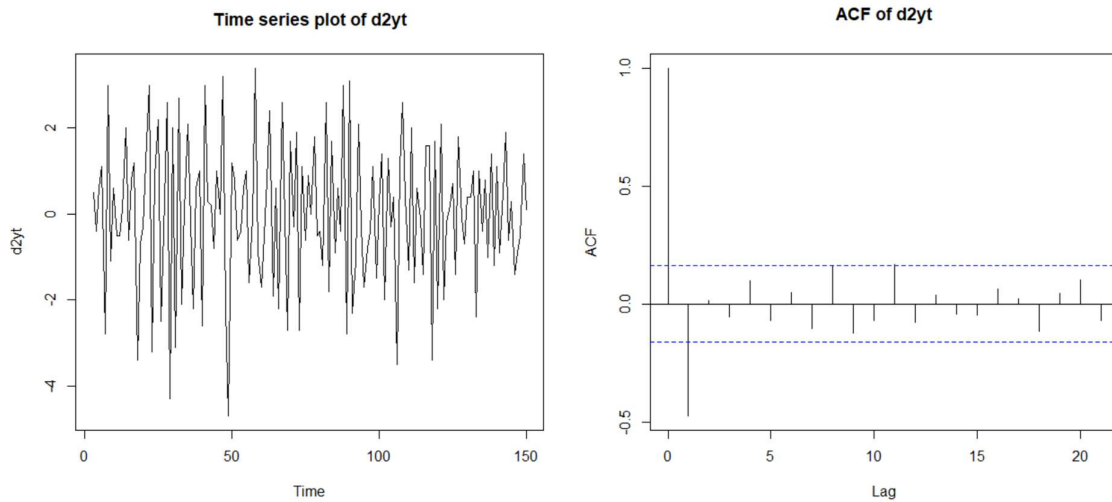
alternative hypothesis: stationary

After 2 differencing

```
d2yt = diff(dyt)
```

```
plot(d2yt,main="Time series plot of d2yt")
```

```
acf(d2yt,main="ACF of d2yt")
```



```
adf.test(d2yt)
```

Augmented Dickey-Fuller Test

data: d2yt

Dickey-Fuller = -6.562, Lag order = 5, p-value = 0.01

alternative hypothesis: stationary

PRACTICAL 9

Code & Output

```
yt=BJsales
for(i in 0:2){
  for(j in 0:2){
    est_i=arima(yt,order=c(i,2,j))
    print(paste('AIC for ARIMA(',i,',2,',j,') is',round(est_i$aic,4))))}
[1] "AIC for ARIMA( 0 ,2, 0 ) is 577.6777"
[1] "AIC for ARIMA( 0 ,2, 1 ) is 517.1371"
[1] "AIC for ARIMA( 0 ,2, 2 ) is 518.9969"
[1] "AIC for ARIMA( 1 ,2, 0 ) is 541.9646"
[1] "AIC for ARIMA( 1 ,2, 1 ) is 518.9734"
[1] "AIC for ARIMA( 1 ,2, 2 ) is 515.579" # best model
[1] "AIC for ARIMA( 2 ,2, 0 ) is 532.2986"
[1] "AIC for ARIMA( 2 ,2, 1 ) is 520.2684"
[1] "AIC for ARIMA( 2 ,2, 2 ) is 521.9854"
```

Best model: ARIMA(1,2,2) as the aic is minimum for ARIMA(1,2,2) model

```
best=arima(yt,order=c(1,2,2))
```

best

Call:

```
arima(x = yt, order = c(1, 2, 2))
```

Coefficients:

```
      ar1      ma1      ma2
0.8598 -1.6254  0.6254
s.e. 0.0796  0.1144  0.1133
```

sigma^2 estimated as 1.766: log likelihood = -253.79, aic = 515.58

Normality of Residuals

```
res = residuals(best)
```

```
qqnorm(res,main="Q-Q plot of residuals")
```

```
qqline(res)
```

```
shapiro.test(res)
```

Shapiro-Wilk normality test

data: res

W = 0.99176, p-value = 0.5385

```
yt_fit=yt-res
```

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Forecast for next 30 observations

```
predict(best,30)$pred
```

Time Series:

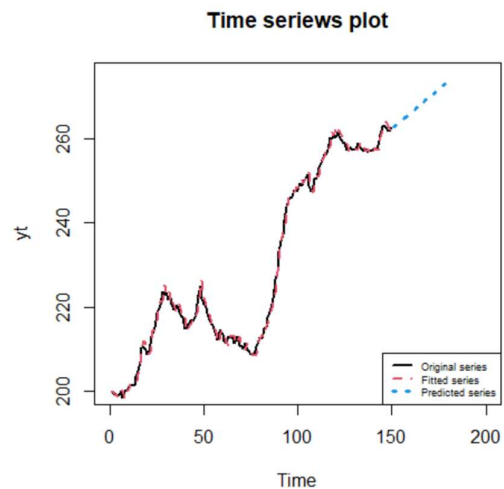
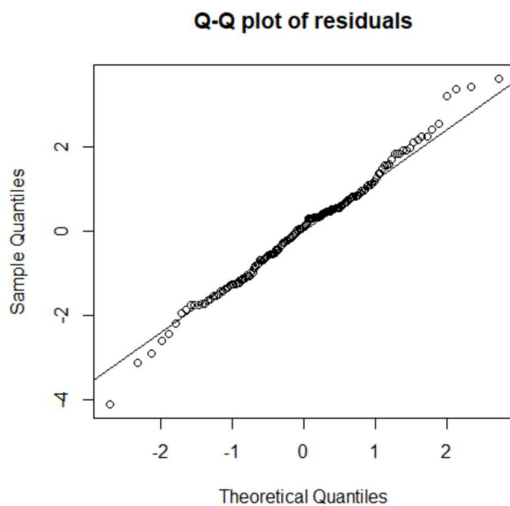
Start = 151

End = 180

Frequency = 1

```
[1] 262.9988 263.3112 263.6354 263.9696 264.3126 264.6630 265.0198 265.3821 265.7491  
266.1203 266.4949 266.8726 267.2528 267.6353 268.0197  
[16] 268.4058 268.7932 269.1819 269.5716 269.9622 270.3536 270.7457 271.1383 271.5315  
271.9250 272.3189 272.7132 273.1077 273.5024 273.8973
```

```
ts.plot(yt,lwd=2,xlim=c(0,200),ylim=c(200,275),main="Time series plot")  
points(yt_fit,type='l',col=2,lty=2,lwd=2)  
points(predict(best,30)$pred,type='l',col=4,lty=3,lwd=3)  
legend('bottomright',legend=c("Original series","Fitted series","Predicted  
series"),lty=c(1,2,3),lwd=c(2,2,3),col=c(1,2,4),cex=0.6)
```



PRACTICAL 10

To estimate the mean survival time and reliability

100 electronic tubes of a certain type were tested, and the test is terminated after first 15 tubes. The failure times occurred after the following hours-

40, 60, 90, 120, 195, 260, 315, 420, 501, 620, 650, 730, 815, 910, 980. Estimate the mean survival time and reliability at $t=500$ hours assuming

i) failed tubes are not replaced

ii) failed tubes are replaced by new tubes

Code & Output

```
# N = 100, r = 15
```

```
ftime = c(40,60,90,120,195,260,350,420,501,620,650,730,815,910,980)
```

```
t = 500
```

```
r = 15
```

```
N = 100
```

Ca se 1

```
lambda1 = (sum(ftime)+(N-r)*max(ftime))/r
```

```
lambda1
```

```
[1] 6002.733
```

```
R1 = exp(-t/lambda1)
```

```
R1
```

```
[1] 0.9200793
```

Case 2

```
lambda2 = N*max(ftime)/r
```

```
lambda2
```

```
[1] 6533.333
```

```
R2 = exp(-t/lambda2)
```

```
R2
```

```
[1] 0.9263246
```

PRACTICAL 11

To estimate the parameter and reliability function

Sixty items were placed on test and the test was terminated after the first 10 items were failed. The failure times (in hours) were recorded as follows:

85,151,280,376,492,520,623,715,820,914

Assuming the failure time distribution is exponential. Estimate the parameter and reliability function at $t = 600$ hours if the failed items are:

- i) Not replaced
- ii) Replaced

Code & Output

N = 60, r = 10

ftime = c(85,151,280,376,492,520,623,715,820,914)

t = 600

r = 10

N = 60

Case 1

$\lambda_1 = (\text{sum}(\text{ftime}) + (N - r) * \max(\text{ftime})) / r$

λ_1

[1] 5067.6

$R_1 = \exp(-t / \lambda_1)$

R_1

[1] 0.8883413

Case 2

$\lambda_2 = N * \max(\text{ftime}) / r$

λ_2

[1] 5484

$R_2 = \exp(-t / \lambda_2)$

R_2

[1] 0.8963636